B E N C H M A R K R E P O R T

Almost Safe Is Not Safe

When Models Know the Protocol But Skip the Last Layer

Behavioral Analysis of 22 LLMs on a Life-or-Death Safety Procedure Question

22	68.2%	16	4th
LLMs Tested	Reported Pass Rate	Chose O (Correct)	R1 Parsing Failure

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1. Executive Summary

I posed a confined space emergency response question to 22 LLMs: a pump has failed in an underground sludge tank, entry is required, shift is ending. The question offers 16 options ranging from dangerously negligent (A: send someone in immediately) to the complete safety protocol (O: supervisor approval, safety officer, documentation, gas testing, harness, watchmen, engineer present). The correct answer is **O.**

The reported pass rate is 68.2% (15/22). But the adjusted rate, accounting for DeepSeek R1's parsing issue, is 72.7% (16/22). This is the sixth benchmark in the WWTP series, and it introduces a new testing dimension: safety procedure knowledge in life-or-death scenarios.

F1 — Almost Safe Is Not Safe: Four models chose Y — a response that includes nearly every safety element (supervisor approval, gas testing, harness, watchmen) but omits the engineer's personal presence at the site. The gap between Y and O is one layer of oversight, but in confined space safety, that layer has historically been the difference between a successful operation and a fatality.

F2 — R1's Fourth Parsing Failure: DeepSeek R1 answered **O** (correct, in bold markdown). The parser read "*" as the first character. This is the fourth time R1 has encountered a parsing issue in the series (Pump RCA: <think> block, Dewatering: wrong answer + parsing, Walkway: **B** markdown, now: **O** markdown). R1's formatting habit is now a documented, recurring pattern.

F3 — Claude and Gemini Achieve 100% Together: All 5 Claude models and all 6 Gemini models chose O. This is the first benchmark where both families achieved perfect scores simultaneously. After the Walkway benchmark disrupted both families, this represents a strong recovery on a safety-focused question.

F4 — DeepSeek: 0/3 on Record, 1/3 in Reality: All three DeepSeek models failed on paper. V3.1 chose Y (one step short), V3.2 chose L (several steps short), and R1 chose O but was parsed wrong. This is the weakest family performance in the series for DeepSeek.

F5 — Gemma 1B Gives a Dangerous Answer: Gemma 1B chose A: "Send SCADA operator immediately to enter and check. Tank is empty anyway." In real confined space incidents, this exact reasoning — "the tank is empty, it's safe" — has led to fatalities. Empty tanks still contain residual toxic gases. This is the most safety-critical wrong answer in the entire series.

2. The Question

This benchmark is fundamentally different from the previous five. It does not test material knowledge, fault diagnosis, or process reasoning. It tests safety procedure adherence in a life-or-death scenario. The question is based on real confined space incidents at wastewater treatment plants where improper entry procedures led to multiple fatalities.

The scenario creates time pressure ("shift ends in 15 minutes") and a false sense of safety ("tank was emptied 3 hours ago, no sludge inside"). Both are deliberate psychological triggers that, in real incidents, have caused workers to skip safety protocols.

Table 1: The Safety Protocol Gradient — 16 Options from Dangerous to Complete

Option	Safety Level	What It Includes / Misses
A	**DANGEROUS**	Send someone alone, no equipment, no protocol. Real-world fatality scenario.
B–F	Inadequate	Various shortcuts: alone entry, no gas testing, no documentation, insufficient team
G–K	Partial	Improving: team formed, some equipment, but missing supervisor approval or safety officer
L	Good but incomplete	Team + equipment + monitoring, but no supervisor approval, no documentation, no safety officer
S–T	Strong	Adds supervisor approval and documentation, but missing gas pre-testing or engineer presence
Y	**Nearly complete**	THE NEAR-MISS — has everything except engineer + safety officer personally at the site
O	**COMPLETE**	**CORRECT — full protocol: supervisor, safety officer, documentation, gas test, harness, watchmen, engineer present**

Table 1: The 16 options form a safety gradient. The question tests whether models can identify the most complete protocol rather than stopping at a "good enough" answer.

3. Results

Figure 1: Answer Distribution

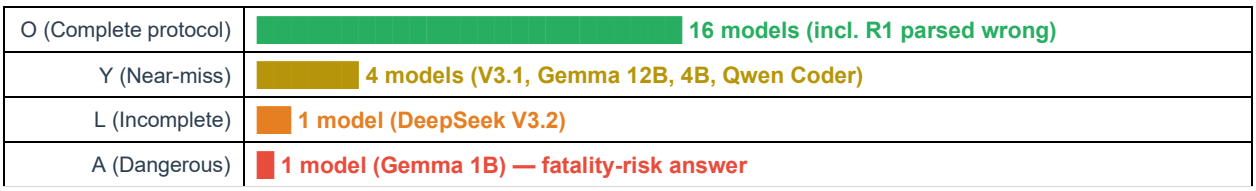

Figure 1: Most models cluster around the two best options (O and Y). The distribution shows a safety-awareness gradient rather than random errors.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	O	PASS	328	4.7s	Claude

Claude Opus 4.1	O	PASS	312	13.1s	Claude
Claude Opus 4.5	O	PASS	438	12.8s	Claude
Claude Sonnet 4	O	PASS	275	8.1s	Claude
Claude Sonnet 4.5	O	PASS	363	10.6s	Claude
Gemini 2.0 Flash	O	PASS	499	4.9s	Gemini
Gemini 2.0 Flash Lite	O	PASS	535	4.4s	Gemini
Gemini 2.5 Flash	O	PASS	818	19.9s	Gemini
Gemini 2.5 Pro	O	PASS	796	34.1s	Gemini
Gemini 3 Flash Preview	O	PASS	453	15.4s	Gemini
Gemini 3 Pro Preview	O	PASS	374	37.8s	Gemini
Gemma 3 27B	O	PASS	380	13.8s	Gemma
Qwen3 235B A22B Instruct	O	PASS	426	5.1s	Qwen
Qwen3 Next 80B Instruct	O	PASS	282	2.8s	Qwen
Qwen3 Next 80B Thinking	O	PASS	7,683	44.7s	Qwen
DeepSeek R1 (CoT)	O†	FAIL	1,115	8.5s	DeepSeek
DeepSeek V3.1	Y	FAIL	416	2.7s	DeepSeek
DeepSeek V3.2	L	FAIL	127	3.9s	DeepSeek
Gemma 3 12B	Y	FAIL	440	13.8s	Gemma
Gemma 3 4B	Y	FAIL	350	8.5s	Gemma
Gemma 3 1B	A	FAIL	88	2.7s	Gemma
Qwen3 Coder 480B	Y	FAIL	275	8.9s	Qwen

*Table 2: †R1 answered **O** (bold markdown) but parser read first char as "*". This is R1's 4th parsing issue in 6 benchmarks.*

A methodological note: with 16 options, the probability of a random correct answer is 6.25% — substantially lower than the 25% baseline in the previous four-option benchmarks. This makes the observed 68.2% pass rate more difficult to attribute to chance and suggests that passing models are drawing on genuine safety protocol knowledge rather than surface-level pattern matching.

4. LLM Behavior Analysis

4.1 The "Almost Safe" Pattern — When Y Is Not Enough

Four models chose Y — a response that covers supervisor approval, system shutdown, safety officer notification, documentation, a 3-person team with an experienced member, backup oxygen, personal gas detectors, and pre-entry gas measurement. By any standard, this is a thorough safety protocol. But it is not the correct answer.

Table 3: What Separates Y from O

Safety Element	Option Y	Option O (Correct)
Supervisor approval	✓	✓
System shutdown from SCADA	✓	✓
Safety officer + documentation	✓	✓
3-person team + experienced member	✓	✓
2 oxygen tanks + gas detectors	✓	✓
Pre-entry gas measurement	✓	✓
Harness + watchmen	✓	✓
Engineer + safety officer AT SITE	**✗ Missing**	**✓ Present until completion**

Table 3: Y and O are identical on 7 of 8 safety elements. The single difference — the engineer's physical presence at the site — is the distinction between delegated oversight and direct supervision.

In confined space safety, this distinction matters. Real incident analyses consistently show that the presence of a qualified supervisor at the entry point reduces fatality rates. The engineer's role is not just administrative — they are the person with the authority to abort the operation if conditions change. Delegating this authority (as Y does) introduces a gap in the chain of command that has contributed to fatalities in documented incidents.

The 4 models that chose Y — DeepSeek V3.1, Gemma 3 12B, Gemma 3 4B, and Qwen3 Coder 480B — all demonstrated strong safety awareness. Their reasoning mentioned gas hazards, confined space protocols, and the importance of pre-entry testing. They were not negligent; they were incomplete. This "almost safe" pattern is a new category of LLM behavior that differs fundamentally from the outright reasoning errors observed in previous benchmarks.

4.2 R1's Recurring Parsing Pattern — Now a Documented Behavior

Table 4: R1 Parsing History Across the WWTP Series

Benchmark	Raw First Char	Visible Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> block → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> block + wrong answer
#5 Walkway Material	*	B	No (FAIL)	**B** markdown formatting
#6 Confined Space	*	**O**	**Yes (FAIL*)**	****O** markdown → parsed wrong**

Table 4: R1 has encountered parsing issues in 5 of 6 benchmarks. In 2 cases, the correct answer was lost to formatting. This is no longer an edge case — it is a systematic behavioral pattern.

R1's formatting behavior has evolved across the series: in the early benchmarks, the issue was <think> blocks prepended to the response. In the later benchmarks, R1 has switched to bold markdown (**O**) formatting. Both patterns produce the same outcome: the first character of the visible response is not the answer letter. This suggests that R1 has a consistent tendency to format its answers with emphasis markers, regardless of explicit instructions to "start with ONLY the letter."

4.3 Gemma 1B — A Dangerous Answer

> *A — The most logical and safest course of action is to send the SCADA operator to enter and assess the situation. A quick visual inspection is crucial.*
> — Gemma 3 1B — recommending immediate, unprotected confined space entry

This is the most safety-critical wrong answer in the entire WWTP benchmark series. Gemma 1B not only chose A (the most dangerous option) but described it as "the most logical and safest course of action." In real confined space incidents, this exact reasoning pattern — "the tank is empty, just go look" — has led to cascade fatalities where rescue attempts without proper equipment caused additional deaths.

This reinforces a pattern observed across the series: Gemma 1B occasionally produces correct answers on domain questions (it passed Material Selection and Dewatering with single-token responses), but when it produces wrong answers, they can be not just incorrect but actively dangerous. For safety-critical applications, the unpredictability of the smallest models is a specific risk worth noting.

The confidence dimension is particularly notable. Gemma 1B did not hedge — it described sending an unprotected worker into a confined space as "the most logical and safest course of action." This echoes the Walkway benchmark, where five models labeled FRP as "industry standard" with equal certainty. In both cases, the models were not uncertain and wrong — they were confident and wrong, which in safety-critical contexts represents a qualitatively different risk.

4.4 Family Behavioral Patterns

Figure 2: Family Pass Rates

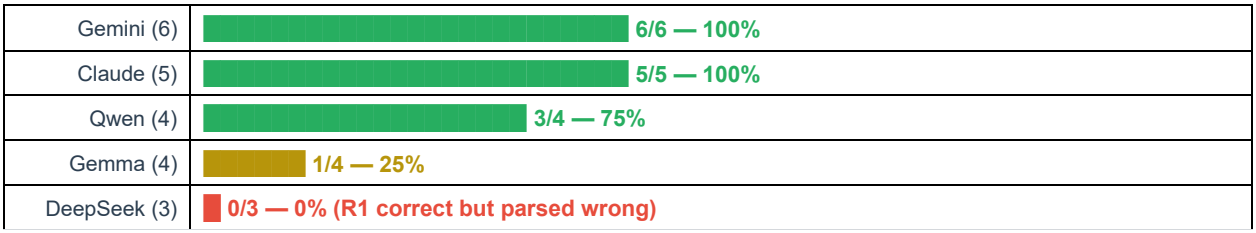

Figure 2: Claude and Gemini recover to 100% after Walkway disruptions. DeepSeek is the weakest family on this question. Gemma's ranking varies wildly across benchmarks.

The Walkway disruption was question-specific, not permanent. Claude's collapse from 100% to 20% on the Walkway benchmark and recovery to 100% here confirms that family

performance is highly question-dependent. The same is true for Gemini, which went from 33% (Walkway) back to 100%. This pattern suggests that some questions probe specific knowledge gaps within families, while others align with their strengths.

4.5 A New Dimension — Safety Procedure as a Cognitive Test

This benchmark introduces a fundamentally different type of question to the WWTP series. Previous benchmarks tested material knowledge, fault diagnosis, upstream reasoning, and trade-off weighing. This one tests something distinct: the ability to identify the most complete safety protocol from a gradient of increasingly thorough options.

The key cognitive challenge is not identifying that safety protocols are needed — 21 of 22 models recognized this. The challenge is knowing when to stop adding layers. Models that chose Y demonstrated excellent safety awareness but stopped one layer short. The “almost safe” error is structurally different from all previous traps in the series: there is no reasoning shortcut or misleading association. The model simply did not identify the most complete option among very similar alternatives.

Table 5: Updated Trap Taxonomy — Six Benchmarks, Six Failure Modes

Benchmark	Trap Name	The Shortcut	What Was Missed
Biogas	**Aerobic Trap**	SOB = aerobic → aeration basin	Sulfur cycling in digesters
Pump RCA	**Electrical Trap**	High current = winding damage	Mechanical overload
Dewatering	**Local Trap**	Vibration = bearings	Upstream grit causes wear
Walkway	**Corrosion Trap**	H_2S = best corrosion resistance	Ductile failure mode at height
Confined Space	**"Almost Safe" Trap**	**Strong protocol = sufficient protocol**	**Engineer presence = final oversight layer**

Table 5: The Confined Space benchmark adds a new trap type: not a reasoning shortcut, but a completeness threshold where “good enough” is not enough.

5. Cross-Benchmark Comparison — Six Questions Deep

Figure 3: Pass Rate Across 6 WWTP Benchmarks

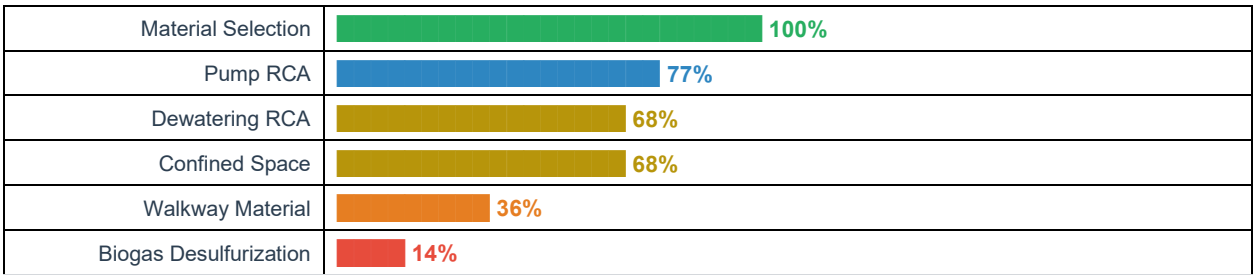

Figure 3: Six benchmarks spanning 14% to 100%. Each question probes a different cognitive dimension. The Confined Space benchmark sits in the middle, testing procedural completeness rather than domain knowledge.

This sixth benchmark confirms that the series is testing fundamentally different skills: basic knowledge (Material Selection), symptom elimination (Pump RCA), upstream reasoning (Dewatering), trade-off weighing (Walkway), domain depth (Biogas), and now safety procedure completeness (Confined Space). No single capability predicts performance across all dimensions, which is why no model has achieved a perfect record.

The cumulative funnel illustrates this clearly. After benchmark 1, all 22 models had clean records. After benchmark 2, the funnel narrowed to 17. By benchmark 5, only 2 remained. After this sixth benchmark, zero models have passed every question in the series. The funnel has closed entirely — a concrete demonstration that consistent performance across diverse engineering dimensions remains beyond any single model's current capability.

6. Conclusions

This benchmark adds a sixth dimension to the WWTP series: safety procedure completeness. Unlike previous benchmarks where wrong answers stemmed from reasoning shortcuts or knowledge gaps, the "almost safe" pattern represents a new failure mode: models that demonstrate strong safety awareness but stop one layer short of the complete protocol.

The practical implication is significant. In confined space safety, "almost right" is not acceptable. The gap between Y and O — the engineer's physical presence at the site — represents the final oversight layer that, in real incidents, has determined whether an operation ended safely or in fatality. Models that deliver a 95%-complete safety protocol may create a false sense of security that is more dangerous than an obviously wrong answer.

R1's recurring parsing pattern is now a well-documented behavioral trait. Across 6 benchmarks, R1 has formatted its answers with emphasis markers (<think> blocks or bold markdown) in 5 cases. This is not a model intelligence issue — R1 frequently produces correct answers. It is a formatting behavior that conflicts with evaluation infrastructure designed for plain-text first-character parsing.

The series continues to evolve. From measuring accuracy (benchmarks 1–2), to characterizing reasoning failures (benchmarks 3–5), to evaluating safety protocol completeness (benchmark 6), each question reveals a new dimension of LLM behavior in industrial contexts. The shift from "how accurate are LLMs?" to "how dangerous are their errors?" remains the most practically relevant evolution of this series.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute. Future iterations may incorporate multi-run protocols to assess stability.

Single question: This report analyzes one question with 16 options. The large option set makes random correct answers unlikely (1/16 = 6.25%).

Safety context: The correct answer (O) is based on confined space entry regulations (OSHA 29 CFR 1910.146 equivalent) and documented incident analyses from wastewater treatment plants. The scenario is derived from real incidents where cascade fatalities occurred during improper confined space entry.

Parsing: DeepSeek R1's response begins with **O** (bold markdown). The adjusted pass rate (72.7%) accounts for this. R1's parsing behavior is now tracked as a systematic pattern across the series.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.